CORE PURE / DIFFERENTIAL EQUATIONS

2nd Order Differential Equations

Mark scheme 12 questions · 125 marks

EXAM BOARDS Edexcel 9FM0 · AQA 7367 · OCR A H245 · OCR MEI H645 · CIE 9231

Questions

Question	Marks by part	Marks	Page
Question 1	(a) 2 (b) 4	6	4
Question 2		11	5
Question 3	(a) 7 (b) 4	11	6
Question 4		11	7
Question 5	(a) 2 (b) 5	7	8
Question 6		11	9
Question 7		11	10
Question 8		11	11
Question 9	(a) 2 (b) 10	12	12
Question 10	(a) 2 (b) 10	12	13
Question 11	(a) 2 (b) 10	12	14
Question 12		10	15
	Total	**125**	

Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

Code	Meaning
M1	Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted.
dM1	Dependent method: as M1, but only if the method mark before it was earned, unless its note names another.
A1	Accuracy: a correct answer or step. It needs the method mark it follows.
A1ft	As A1, following through from an earlier answer that the note names.
A1 AG	The last mark of a part whose answer is printed in the question: complete working, with no errors.
B1	A correct value, fact or statement. It needs no method mark.
B1ft	As B1, following through from an earlier answer that the note names.
E1	A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark.

Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes. A grey line gives the rule where the name alone doesn't.

ft	follow through Correct work from the student's own earlier wrong value earns the mark.	**AG**	answer given The answer is printed in the question, so every step to it must be shown.
cao	correct answer only No follow-through from an earlier error.	**cso**	correct solution only No error anywhere in the part, not just in its answer.
oe	or equivalent	**awrt**	answers which round to
soi	seen or implied Earned if written, or if later working shows it was used.	**isw**	ignore subsequent working A correct answer, once reached, isn't spoiled by later working.
SC	special case Credit, as the note says, for work outside the main method.	**their**	the student's own earlier value
condone	a slip that doesn't cost the mark	[ ]	working the mark doesn't need
s.f.	significant figures	**d.p.**	decimal places

Reading the scheme

- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- [$\boldsymbol{n}$] ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

Accuracy and answers

- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

Value of g

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state g, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

Question	Scheme	Marks	Notes
1(a)	$\lambda^2 + 4 = 0 \Rightarrow \lambda = \ldots$	**M1**	Forms the auxiliary equation and solves it by any method.
	$\lambda = \pm 2i$ $(y =)\ C_1 \cos 2x + C_2 \sin 2x$	**A1**	Must be in terms of $\cos 2x$ and $\sin 2x$, oe such as $R\cos(2x + \alpha)$.
		[2]	
1(b)	$y = x(A\cos 2x + B\sin 2x)$ $y' = A\cos 2x + B\sin 2x$ $\quad + x(-2A\sin 2x + 2B\cos 2x)$ $y'' = -4A\sin 2x + 4B\cos 2x$ $\quad - 4x(A\cos 2x + B\sin 2x)$ $4B\cos 2x - 4A\sin 2x = 8\cos 2x - 12\sin 2x$ $\Rightarrow A = \ldots,\ B = \ldots$	**M1**	Differentiates the given form twice by the product rule, substitutes into the equation and equates coefficients of $\cos 2x$ and $\sin 2x$. Condone slips in differentiation.
	$A = 3$ or $B = 2$	**A1**	
	$x(3\cos 2x + 2\sin 2x)$	**A1**	Both constants correct, in the particular integral or stated.
	$(y =)\ C_1 \cos 2x + C_2 \sin 2x$ $\quad + x(3\cos 2x + 2\sin 2x)$	**A1ft**	ft their complementary function from (a) plus their particular integral of the form $x(p\cos 2x + q\sin 2x)$ with numerical p, q.
		[4]	
	Total for Question 1	**6**	

Question	Scheme	Marks	Notes
2	$\lambda^2 + 4\lambda + 5 = 0 \Rightarrow \lambda = \ldots$	**M1**	Forms the auxiliary equation and solves it by formula, completing the square or calculator.
	$\lambda = -2 \pm i$ $(y =)\ e^{-2t}(A\cos t + B\sin t)$	**A1**	
	$y = a\cos t + b\sin t$	**B1**	
	$\frac{\mathrm{d}y}{\mathrm{d}t} = -a\sin t + b\cos t,\ \frac{\mathrm{d}^2y}{\mathrm{d}t^2} = -a\cos t - b\sin t$ $(-a + 4b + 5a)\cos t + (-b - 4a + 5b)\sin t$ $= 8\cos t$	**M1**	Differentiates their trial function twice, substitutes into the equation and equates coefficients of $\cos t$ and $\sin t$. Condone sign slips in the derivatives.
	$4a + 4b = 8, \quad -4a + 4b = 0$	**A1**	oe
	$\Rightarrow a = \ldots,\ b = \ldots$	**dM1**	Solves their two linear equations for both a and b, by any means.
	$(y =)\ \cos t + \sin t$	**A1**	
	$y = e^{-2t}(A\cos t + B\sin t) + \cos t + \sin t$	**A1ft**	ft their complementary function with two arbitrary constants plus their particular integral; must be $y = \ldots$ in terms of t.
	$t = 0,\ y = 3:\ 3 = A + 1 \Rightarrow A = \ldots$	**M1**	Substitutes $t = 0$, $y = 3$ into their general solution with two arbitrary constants.
	$\frac{\mathrm{d}y}{\mathrm{d}t} = -2e^{-2t}(A\cos t + B\sin t)$ $+ e^{-2t}(-A\sin t + B\cos t) - \sin t + \cos t$ $t = 0:\ 1 = -2A + B + 1 \Rightarrow B = \ldots$	**M1**	Differentiates their general solution, using the product rule on the exponential term, substitutes $t = 0$, $\frac{\mathrm{d}y}{\mathrm{d}t} = 1$ and solves for both constants. Condone sign slips.
	$y = e^{-2t}(2\cos t + 4\sin t) + \cos t + \sin t$	**A1**	Must be stated as $y = \ldots$ in terms of t; $A = 2$, $B = 4$ alone is not enough.
	Total for Question 2	**11**	

Question	Scheme	Marks	Notes
3(a)	$\lambda^2 + 6\lambda + 13 = 0 \Rightarrow \lambda = \ldots$	**M1**	Forms the auxiliary equation and solves it by formula, completing the square or calculator, reaching complex roots.
	$\lambda = -3 \pm 2i$ $(y =)\, e^{-3t}(A\cos 2t + B\sin 2t)$	**A1**	Correct complementary function oe; roots alone are not enough.
	$y = a\sin 2t + b\cos 2t$	**B1**	
	$-4a\sin 2t - 4b\cos 2t + 6(2a\cos 2t - 2b\sin 2t)$ $\quad + 13(a\sin 2t + b\cos 2t) = 75\sin 2t$ Equates coefficients of $\sin 2t$ and $\cos 2t$	**M1**	Differentiates their trial function twice, substitutes into the equation and equates coefficients of $\sin 2t$ and $\cos 2t$. Condone sign and coefficient slips in the derivatives.
	$9a - 12b = 75, \quad 12a + 9b = 0$	**A1**	oe
	$a = 3,\ b = -4$ $(y =)\, 3\sin 2t - 4\cos 2t$	**A1**	
	$y = e^{-3t}(A\cos 2t + B\sin 2t) + 3\sin 2t - 4\cos 2t$	**A1ft**	ft their complementary function with two arbitrary constants plus their particular integral with none. Needs $y =$.
		[7]	
3(b)	$t \to \infty \Rightarrow e^{-3t}(A\cos 2t + B\sin 2t) \to 0$ for all A, B, so $y \approx 3\sin 2t - 4\cos 2t$	**E1**	States that $e^{-3t} \to 0$, so the complementary function vanishes whatever A and B (or the initial conditions), leaving the particular integral.
	$R\sin(2t - \phi) = R\cos\phi\sin 2t - R\sin\phi\cos 2t$ $R\cos\phi = 3, \quad R\sin\phi = 4$	**M1**	Expands $R\sin(2t - \phi)$ and equates coefficients with their particular integral of the form $p\sin 2t + q\cos 2t$, or uses $R = \sqrt{p^2 + q^2}$ and $\tan\phi$ with their p, q. Condone sign slips.
	$R = 5$	**A1**	
	$\phi = \arctan\frac{4}{3}$ $[\approx 0.927]$	**A1**	$\arctan\frac{4}{3}$ oe, such as $\arcsin\frac{4}{5}$, or awrt 0.927. Not in degrees.
		[4]	
	Total for Question 3	**11**	

Question	Scheme	Marks	Notes
4	$\lambda^2 + 2\lambda + 5 = 0 \Rightarrow \lambda = \ldots$	**M1**	Forms the auxiliary equation and solves it by formula, completing the square or calculator, leading to $\lambda = \ldots$.
	$\lambda = -1 \pm 2i$ $(y =)\ e^{-t}(A\cos 2t + B\sin 2t)$	**A1**	Any equivalent real form, such as $Re^{-t}\cos(2t + \alpha)$.
	$(y =)\ a\cos t + b\sin t$	**B1**	
	$(-a\cos t - b\sin t) + 2(-a\sin t + b\cos t)$ $+ 5(a\cos t + b\sin t) = 10\cos t$	**M1**	Differentiates their trial function twice, substitutes into the equation and equates coefficients of $\cos t$ and $\sin t$. Condone sign slips in the derivatives.
	$4a + 2b = 10, \quad 4b - 2a = 0$	**A1**	oe
	$\Rightarrow a = \ldots,\ b = \ldots$	**dM1**	Solves their two simultaneous equations for a and b by any means.
	$a = 2,\ b = 1$ $(y =)\ 2\cos t + \sin t$	**A1**	
	$y = e^{-t}(A\cos 2t + B\sin 2t) + 2\cos t + \sin t$	**A1ft**	ft their complementary function, with two arbitrary constants, plus their particular integral. Needs $y =$.
	$5 = A + 2 \Rightarrow A = \ldots\ [= 3]$	**M1**	Substitutes $t = 0$, $y = 5$ into their general solution with two arbitrary constants to find one constant.
	$\frac{\mathrm{d}y}{\mathrm{d}t} = -e^{-t}(A\cos 2t + B\sin 2t)$ $+ e^{-t}(-2A\sin 2t + 2B\cos 2t)$ $- 2\sin t + \cos t$ $2 = -\text{“3”} + 2B + 1 \Rightarrow B = \ldots\ [= 2]$	**M1**	Differentiates their general solution, using the product rule on any product, substitutes $t = 0$, $\frac{\mathrm{d}y}{\mathrm{d}t} = 2$ and solves for the other constant. Condone sign slips.
	$y = e^{-t}(3\cos 2t + 2\sin 2t) + 2\cos t + \sin t$	**A1**	cso. Needs $y =$.
	Total for Question 4	**11**	

Question	Scheme	Marks	Notes
5(a)	$\lambda^2 + 4\lambda + 13 = 0 \Rightarrow \lambda = \ldots$	**M1**	Forms the auxiliary equation and solves it by formula, completing the square or calculator, reaching $\lambda = \ldots$.
	$\lambda = -2 \pm 3i$ $y = e^{-2x}(A\cos 3x + B\sin 3x)$	**A1**	Real form, oe.
		[2]	
5(b)	$y = e^{-2x}(a\cos x + b\sin x)$	**B1**	
	$\frac{\mathrm{d}y}{\mathrm{d}x} = e^{-2x}[(b-2a)\cos x - (a+2b)\sin x]$ $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = e^{-2x}[(3a-4b)\cos x + (4a+3b)\sin x]$ $\Rightarrow 8a\cos x + 8b\sin x = 6\cos x - 8\sin x$ $\Rightarrow a = \ldots,\ b = \ldots$	**M1**	Differentiates their trial form twice using the product rule, substitutes into the equation and equates coefficients of $\cos x$ and $\sin x$. Condone sign and coefficient slips.
	$a = \frac{3}{4}$ or $b = -1$	**A1**	Either constant correct.
	$(y =)\ e^{-2x}\left(\frac{3}{4}\cos x - \sin x\right)$	**A1**	
	$y = e^{-2x}\left(A\cos 3x + B\sin 3x + \frac{3}{4}\cos x - \sin x\right)$	**A1ft**	ft their complementary function from (a), with two arbitrary constants, plus their particular integral; must start $y =$.
		[5]	
Total for Question 5		**7**	

Question	Scheme	Marks	Notes
6	$\lambda^2 + 4\lambda + 13 = 0 \Rightarrow \lambda = \ldots$	**M1**	Forms the auxiliary equation and solves it by formula with their values, completing the square or calculator, leading to $\lambda = \ldots$
	$\lambda = -2 \pm 3i$ $y = e^{-2t}(A\cos 3t + B\sin 3t)$	**A1**	Any equivalent real form; the roots alone are not enough.
	$(y =)\ a\cos 3t + b\sin 3t$	**B1**	
	$\frac{\mathrm{d}y}{\mathrm{d}t} = -3a\sin 3t + 3b\cos 3t$ $\frac{\mathrm{d}^2y}{\mathrm{d}t^2} = -9a\cos 3t - 9b\sin 3t$ $(-9a + 12b + 13a)\cos 3t$ $+ (-9b - 12a + 13b)\sin 3t$ $= 28\cos 3t - 4\sin 3t$	**M1**	Differentiates their trial function twice, substitutes all three terms and equates coefficients of $\cos 3t$ and $\sin 3t$. Condone sign and coefficient slips.
	$4a + 12b = 28,\ 4b - 12a = -4$	**A1**	oe
	$a = \ldots,\ b = \ldots$	**dM1**	Solves their two equations simultaneously by any means, calculator included, to find both a and b.
	$y = \cos 3t + 2\sin 3t$	**A1**	
	$y = e^{-2t}(A\cos 3t + B\sin 3t) + \cos 3t + 2\sin 3t$	**A1ft**	ft their complementary function, with two arbitrary constants, plus their particular integral with no arbitrary constants.
	$5 = A + 1\ [\Rightarrow A = 4]$	**M1**	Substitutes $t = 0$, $y = 5$ into their general solution, with two arbitrary constants, to form an equation in the constants.
	$\frac{\mathrm{d}y}{\mathrm{d}t} = -2e^{-2t}(A\cos 3t + B\sin 3t)$ $+ e^{-2t}(3B\cos 3t - 3A\sin 3t)$ $- 3\sin 3t + 6\cos 3t$ $7 = -2A + 3B + 6 \Rightarrow B = \ldots$	**M1**	Differentiates their general solution, with the product rule on the complementary function, substitutes $t = 0$, $\frac{\mathrm{d}y}{\mathrm{d}t} = 7$ and solves for both constants. Condone sign slips.
	$y = e^{-2t}(4\cos 3t + 3\sin 3t) + \cos 3t + 2\sin 3t$	**A1**	Needs $y = \ldots$; $A = 4$, $B = 3$ alone is not enough.
	Total for Question 6	**11**	

Question	Scheme	Marks	Notes
7	$\lambda^2 - \lambda - 6 = 0 \Rightarrow \lambda = \ldots$	**M1**	Forms the auxiliary equation and solves it: factors whose constants multiply to their constant term, the formula, completing the square or a calculator.
	$\lambda = 3,\ -2$ $(y =)\ Ae^{3x} + Be^{-2x}$	**A1**	
	$(y =)\ a\sin x + b\cos x$	**B1**	
	$y' = a\cos x - b\sin x,\ y'' = -a\sin x - b\cos x$ $(-a\sin x - b\cos x) - (a\cos x - b\sin x)$ $-6(a\sin x + b\cos x) = 50\sin x$	**M1**	Differentiates a particular integral of the form $a\sin x + b\cos x$ twice, substitutes into the equation and equates coefficients of $\sin x$ and $\cos x$. Condone sign slips.
	$b - 7a = 50,\quad -a - 7b = 0$	**A1**	oe
	$\Rightarrow a = \ldots,\ b = \ldots$	**dM1**	Solves their two simultaneous equations for a and b, by any means including a calculator.
	$(y =)\ -7\sin x + \cos x$	**A1**	
	$y = Ae^{3x} + Be^{-2x} - 7\sin x + \cos x$	**A1ft**	ft their complementary function plus their particular integral; must be $y = \ldots$ in terms of x.
	$3 = A + B + 1$	**M1**	Substitutes $x = 0$, $y = 3$ into their general solution, of the form complementary function with two arbitrary constants plus particular integral.
	$y' = 3Ae^{3x} - 2Be^{-2x} - 7\cos x - \sin x$ $-6 = 3A - 2B - 7 \Rightarrow A = \ldots,\ B = \ldots$	**M1**	Differentiates their general solution, substitutes $x = 0$, $y' = -6$, and solves with their first equation for A and B. Condone slips in differentiation.
	$y = e^{3x} + e^{-2x} - 7\sin x + \cos x$	**A1**	Must be $y = \ldots$.
	Total for Question 7	**11**	

Question	Scheme	Marks	Notes
8	$\lambda^2 + 4\lambda + 13 = 0 \Rightarrow \lambda = \ldots$	**M1**	Forms the auxiliary equation and solves it by the formula with their values, completing the square or calculator.
	$\lambda = -2 \pm 3i$ $(y =)\ e^{-2t}(A\cos 3t + B\sin 3t)$	**A1**	
	$y = e^{-2t}(a\sin t + b\cos t)$	**B1**	
	$\frac{\mathrm{d}y}{\mathrm{d}t} = -2e^{-2t}(a\sin t + b\cos t)$ $\quad + e^{-2t}(a\cos t - b\sin t)$ $\frac{\mathrm{d}^2y}{\mathrm{d}t^2} = \ldots$	**M1**	Differentiates their particular integral twice, using the product rule on e^{-2t} times the trigonometric terms. Condone sign and coefficient slips.
	$\frac{\mathrm{d}y}{\mathrm{d}t} = e^{-2t}[(-2a - b)\sin t + (a - 2b)\cos t]$ $\frac{\mathrm{d}^2y}{\mathrm{d}t^2} = e^{-2t}[(3a + 4b)\sin t + (3b - 4a)\cos t]$	**A1**	Both derivatives correct; need not be simplified.
	$e^{-2t}(8a\sin t + 8b\cos t) = 8e^{-2t}\sin t$ $\Rightarrow a = \ldots,\ b = \ldots$	**dM1**	Substitutes their derivatives into the equation and equates coefficients of $\sin t$ and $\cos t$ to find a and b. Condone slips.
	$(y =)\ e^{-2t}\sin t$	**A1**	
	$y = e^{-2t}(A\cos 3t + B\sin 3t + \sin t)$	**A1ft**	ft their complementary function with two arbitrary constants plus their particular integral, which has none. Needs $y =$.
	$t = 0,\ y = 1 \Rightarrow A = \ldots\ [= 1]$	**M1**	Substitutes $t = 0$, $y = 1$ into their general solution, complementary function plus particular integral, to find A.
	$\frac{\mathrm{d}y}{\mathrm{d}t} = -2e^{-2t}(A\cos 3t + B\sin 3t + \sin t)$ $\quad + e^{-2t}(-3A\sin 3t + 3B\cos 3t + \cos t)$ $2 = -2\text{“}A\text{”} + 3B + 1 \Rightarrow B = \ldots\ [= 1]$	**M1**	Differentiates their general solution using the product rule, substitutes $t = 0$, $\frac{\mathrm{d}y}{\mathrm{d}t} = 2$ and solves for B with their A. Condone sign slips.
	$y = e^{-2t}(\cos 3t + \sin 3t + \sin t)$	**A1**	cao, with $y =$.
	Total for Question 8	**11**	

Question	Scheme	Marks	Notes
9(a)	$(1+\tanh x)\cosh x = \cosh x + \sinh x$ $= \frac{e^x+e^{-x}}{2} + \frac{e^x-e^{-x}}{2}$	**M1**	Uses $\tanh x = \frac{\sinh x}{\cosh x}$ and exponential definitions, or writes $\tanh x$ and $\cosh x$ in exponentials directly. Condone a sign slip.
	$= e^x$	**A1 AG**	Correct exponential expression leading without error to e^x; combined steps accepted.
		[2]	
9(b)	$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - 2\frac{\mathrm{d}y}{\mathrm{d}x} + y = 3e^x$	**B1**	
	$\lambda^2 - 2\lambda + 1 = 0 \Rightarrow \lambda = \ldots$	**M1**	Forms the auxiliary equation and solves it by any means, leading to $\lambda = \ldots$.
	$(y =)(A + Bx)e^x$	**A1**	
	$y = ax^2e^x$	**B1**	Or any form containing an x^2e^x term, such as $(ax^2 + bx + c)e^x$.
	$y' = e^x(ax^2 + 2ax)$, $y'' = e^x(ax^2 + 4ax + 2a)$ $\Rightarrow 2ae^x = 3e^x \Rightarrow a = \ldots$	**M1**	Differentiates their particular integral twice, substitutes into the differential equation with their right-hand side and solves for a. Condone slips in differentiation.
	$a = \frac{3}{2}$	**A1**	Or $y = \frac{3}{2}x^2e^x$.
	$y = (A + Bx)e^x + \frac{3}{2}x^2e^x$	**A1ft**	ft their complementary function plus their particular integral, with two arbitrary constants.
	$x = 0,\ y = 1 \Rightarrow A = \ldots\ [= 1]$	**M1**	Substitutes $x = 0$, $y = 1$ into their general solution containing two arbitrary constants.
	$\frac{\mathrm{d}y}{\mathrm{d}x} = e^x\left(A + Bx + \frac{3}{2}x^2\right) + e^x(B + 3x)$ $\Rightarrow 0 = A + B \Rightarrow B = \ldots$	**M1**	Differentiates their general solution, product rule on at least one term, substitutes $x = 0$, $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$ and solves for the constants.
	$y = e^x\left(\frac{3}{2}x^2 - x + 1\right)$	**A1**	oe, as y in terms of x.
		[10]	
	Total for Question 9	**12**	

Question	Scheme	Marks	Notes
10(a)	$\frac{\frac{1}{2}(e^x+e^{-x})-\frac{1}{2}(e^x-e^{-x})}{\frac{1}{2}(e^x+e^{-x})+\frac{1}{2}(e^x-e^{-x})}$	**M1**	Replaces $\cosh x$ and $\sinh x$ by their exponential definitions in both numerator and denominator.
	$=\dfrac{e^{-x}}{e^x}=e^{-2x}$	**A1 AG**	Correct substituted expression simplified without error to the printed result; combined steps accepted.
		[2]	
10(b)	$\frac{\mathrm{d}^2y}{\mathrm{d}x^2}+4\frac{\mathrm{d}y}{\mathrm{d}x}+4y=4e^{-2x}$	**B1**	May be implied by working with right-hand side $4e^{-2x}$.
	$\lambda^2+4\lambda+4=0\Rightarrow\lambda=\ldots$	**M1**	Forms the auxiliary equation and solves it by any method.
	$(y=)(A+Bx)e^{-2x}$	**A1**	
	$(y=)\,ax^2e^{-2x}$	**B1**	Or $(ax^2+bx+c)e^{-2x}$.
	$\frac{\mathrm{d}y}{\mathrm{d}x}=(2ax-2ax^2)e^{-2x}$ $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}=(2a-8ax+4ax^2)e^{-2x}$ $\Rightarrow 2ae^{-2x}=4e^{-2x}\Rightarrow a=\ldots$	**M1**	Differentiates their particular integral of the form ax^2e^{-2x} twice by the product rule, substitutes into the equation and solves for a. Condone coefficient slips.
	$(y=)\,2x^2e^{-2x}$	**A1**	
	$y=(A+Bx)e^{-2x}+2x^2e^{-2x}$	**A1ft**	ft their complementary function with two arbitrary constants plus their non-zero particular integral.
	$x=0,\ y=1\Rightarrow A=\ldots\ [=1]$	**M1**	Substitutes $x=0$, $y=1$ into their general solution with two arbitrary constants.
	$\frac{\mathrm{d}y}{\mathrm{d}x}=-2e^{-2x}(2x^2+Bx+A)+e^{-2x}(4x+B)$ $1=B-2A\Rightarrow B=\ldots$	**M1**	Differentiates their general solution, using the product rule on product terms, substitutes $x=0$, $\frac{\mathrm{d}y}{\mathrm{d}x}=1$ and solves for the constants. Condone sign slips.
	$y=e^{-2x}(2x^2+3x+1)$	**A1**	oe.
		[10]	
	Total for Question 10	**12**	

Question	Scheme	Marks	Notes
11(a)	$3\cosh x - \sinh x = \frac{3(e^x+e^{-x})}{2} - \frac{e^x-e^{-x}}{2}$	**M1**	Substitutes exponential definitions of both $\cosh x$ and $\sinh x$. Condone one sign slip.
	$= \frac{2e^x+4e^{-x}}{2} = e^x + 2e^{-x}$	**A1 AG**	Correct substituted expression simplified without error to the printed result; combined steps accepted.
		[2]	
11(b)	$\lambda^2 - \lambda - 2 = 0 \Rightarrow \lambda = \ldots$	**M1**	Forms the auxiliary equation and solves it: factorising with product of constants -2, formula or completing the square; calculator accepted.
	$\lambda = 2,\ -1$ $(y =)\, Ae^{2x} + Be^{-x}$	**A1**	
	$(y =)\, ae^x + bxe^{-x}$	**B1**	Accept with an additional ce^{-x} term.
	$y' = ae^x + b(1-x)e^{-x}$, $y'' = ae^x + b(x-2)e^{-x}$ $-2ae^x - 3be^{-x} = 6e^x + 12e^{-x}$ $\Rightarrow a = \ldots,\ b = \ldots$	**M1**	Differentiates their particular integral, with an e^x and an e^{-x} term, twice, substitutes into the equation and equates coefficients. Condone differentiation slips.
	$a = -3,\ b = -4$	**A1 A1**	
	$y = Ae^{2x} + Be^{-x} - 3e^x - 4xe^{-x}$	**A1ft**	ft their complementary function plus their particular integral; must be $y =$ a function of x.
	$1 = A + B - 3$	**M1**	Substitutes $x = 0$, $y = 1$ into their general solution with two arbitrary constants.
	$y' = 2Ae^{2x} - Be^{-x} - 3e^x - 4(1-x)e^{-x}$ $1 = 2A - B - 7 \Rightarrow A = \ldots,\ B = \ldots$	**M1**	Differentiates their general solution, substitutes $x = 0$, $\frac{dy}{dx} = 1$ and solves both equations for A and B. Condone slips differentiating xe^{-x}.
	$y = 4e^{2x} - 3e^x - 4xe^{-x}$	**A1**	cao; needs $y =$. Condone $+0e^{-x}$ left in.
		[10]	
	Total for Question 11	**12**	

Question	Scheme	Marks	Notes
12	$\lambda^2 - 4\lambda + 4 = 0 \Rightarrow \lambda = \ldots$	**M1**	Forms the auxiliary equation and solves it by any method.
	$\lambda = 2$ (repeated) $(y =)(A + Bx)e^{2x}$	**A1**	
	$y = e^{2x}(ax^3 + bx^2)$	**B1**	Also accept $e^{2x}(ax^3 + bx^2 + cx + d)$.
	$y' = e^{2x}(2ax^3 + (3a + 2b)x^2 + 2bx)$ $y'' = e^{2x}(4ax^3 + (12a + 4b)x^2$ $\quad + (6a + 8b)x + 2b)$ $\Rightarrow e^{2x}(6ax + 2b) = xe^{2x}$ $\Rightarrow 6a = 1,\ 2b = 0$	**M1 A1**	**M1** Differentiates their trial function of the form $e^{2x} \times$ polynomial twice by the product rule, substitutes and equates coefficients. Condone slips in differentiation. **A1** Correct reduced form, or $6a = 1$, $2b = 0$ seen.
	$y = \frac{1}{6}x^3e^{2x}$	**A1**	
	$y = (A + Bx)e^{2x} + \frac{1}{6}x^3e^{2x}$	**A1ft**	ft their complementary function with two arbitrary constants plus their nonzero particular integral.
	$x = 0,\ y = 1 \Rightarrow A = 1$	**M1**	Substitutes $x = 0$, $y = 1$ into their general solution with two arbitrary constants.
	$\frac{\mathrm{d}y}{\mathrm{d}x} = 2e^{2x}\left(A + Bx + \frac{x^3}{6}\right) + e^{2x}\left(B + \frac{x^2}{2}\right)$ $0 = 2A + B \Rightarrow B = \ldots$	**M1**	Differentiates their general solution, product rule on at least one product, substitutes $x = 0$, $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$ and solves for both constants.
	$y = e^{2x}\left(\frac{x^3}{6} - 2x + 1\right)$	**A1**	oe, with $y =$.
	Total for Question 12	**10**	